Auto Feeder Assembly Guide

A step-by-step manual for assembling the single and dual-channel fish feeders

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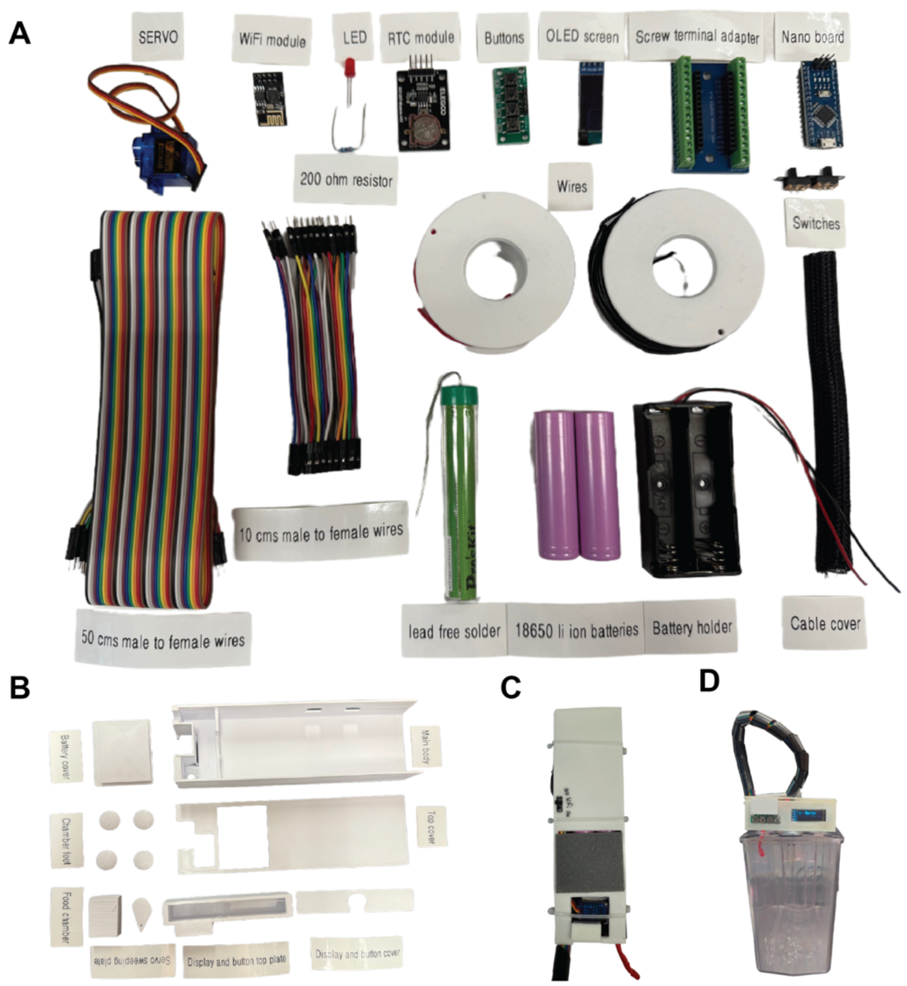
1. Introduction

This document provides a detailed, beginner-friendly guide to assembling an **automated fish feeder** inspired by the NodeMCU-based design by McKay. Our updated version:

- Uses an **Arduino Nano** microcontroller, known for its simplicity and support.
- Features a **fully enclosed** waterproof casing to protect electronics from the humid, splash-prone lab environment.
- Runs on **battery power** (using 18650 cells), avoiding tangled external wires.
- Includes a **screen** (OLED) to display time, status, and feed mode.
- Provides **buttons** for manual control and **LED indicators** for feedback.

This setup ensures consistent feeding times, minimal human error, and supports high-throughput behavioural research with killifish and other aquatic models. The system supports both **single- and dual-feeder configurations**.



2. Materials and Tools Required

2.1 Electronic Components

Verify that you have everything on this list before beginning construction.

Component	Quantity	Notes
Arduino Nano	1	With USB cable
Screw Terminal Adapter (Nano Dock)	1	For easy connections
SG90 Servo Motors + Screws for arm (size)	1 (or 2)	1 = single feeder, 2 = dual
RTC Module (DS1307)	1	Real Time Clock for time-based feeding
OLED Display (SSD1306, 128x32)	1	For showing time and status
Push Buttons	4	For mode switching and triggering feed
LED + Resistor (200Ω)	1 (or 2)	Long leg = positive (+); resistor protects from overvoltage
18650 Li-ion Batteries (3000mAh)	2	Use in **series**
18650 Battery Holder (2-slot)	1	Wired in **series**
Slide Switch	1	For power control
Jumper Wires	Many	Use long wires (e.g. 50cm) for front-mounted components (screen, buttons, LED). Shorter ones for rest. Female ends are helpful for component pins (OLED, Buttons, RTC)
*CAD-printed Enclosure	1	Holds and protects the components

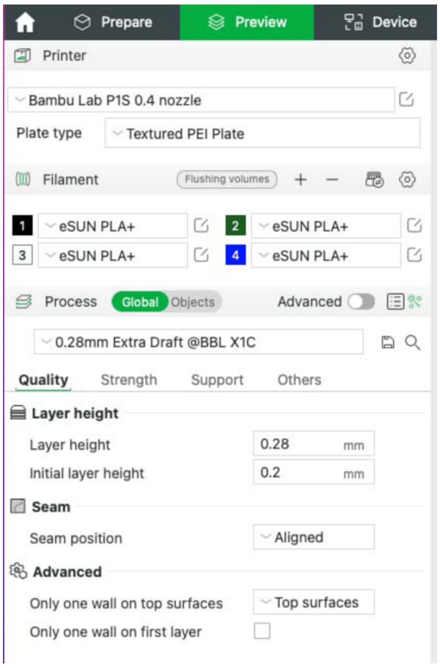
Optional (for Wi-Fi upgrade):

- ESP8266 or ESP32 microcontroller
- Additional resistors and a toggle switch

2.2 Tools Required

- Soldering Iron + Solder
- Wire Cutter and Stripper
- Small Screwdriver (for terminal blocks)
- Tweezers
- Double-sided tape or Hot Glue Gun
- Heat Shrink Tubing or Electrical Tape

***CAD-Printed Enclosure - Download CAD files (STL format)**:*
https://github.com/teesha902/FYP_ASMLab/tree/dev-teesha/Autofeeder/CAD_files



Printing Notes (Tested on Bambu Lab with PLA+):
- **Filament:** 1.75mm PLA+ (Bambu Lab White, 1KG)
- **Print temperature:** 205–225°C
- **Layer height:** 0.2 mm recommended
- **Supports:** Optional (recommended for button holes and battery slot overhangs)
- **Print orientation:** Base down, front panel facing upward
- **Infill:** 15–30% for a sturdy structure
- **Cooling:** Enabled

Ensure screw holes, screen cutouts, and wire passthroughs are clear before mounting electronics. Test fit components before final assembly.

3. Step-by-Step Assembly

3.1 Preliminary Steps

• Gather all components and tools
• It is helpful to organize your wires by color and length (e.g., red/orange = VCC, black/brown = GND)
• Keep the wiring diagram nearby (see attached image)
• Ensure a clean, ventilated workspace (especially if soldering)

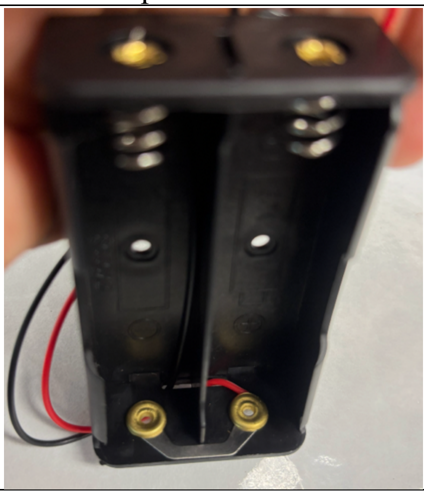
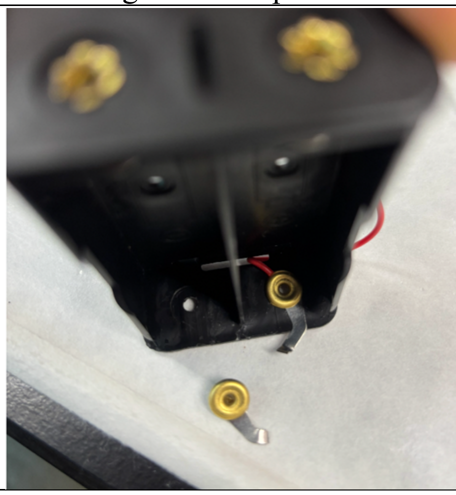
Notes before starting:
- Assembly order is flexible – aim to keep wires untangled
- Dashed wires in the diagram are **optional** (for dual-feeder only)
- Cut and strip all servo and casing wires to ~2 cm (except orange servo wire and 3 cm for switches)
- Strip about **0.5 cm** at wire ends for insertion into holders
- For holders: use 12 cm stripped wire or male ends — tighten firmly and test
- Use female jumper wires for connecting RTC, OLED, and buttons to Arduino. Secure with tape or hot glue
- Attach resistor to long leg (anode) of the LED. Be gentle — LED legs break easily
- **LEDs can only handle 3.3V** — always use a resistor to drop the 5V input

Battery Holder Fix (if received with incorrect parallel wiring):

Follow these steps to fix it:

1. **Cut** the **metal plate connection** on the end where the red and black wires exit the battery holder. This breaks the parallel circuit.
2. **Pull out the black wire** — you'll notice it's attached internally to the opposite terminal (across from the red wire).
3. **Reattach the black wire** to the **same end of the case as the red wire**, but to the **opposite battery slot's terminal**.
   - Use **tweezers** to carefully bend and manipulate the internal metal tab for a firm connection.
   - **Solder** if necessary for a solid contact.

Ensure you now get ~7.4V across the terminals when both batteries are inserted (with correct polarity).

BEFORE	AFTER
Before metal plate connection was cut	After cutting the metal plate connection
	
Old black wire attachment (on wrong end)	Fixed black wire attachment
	

3.2 Mechanical Mounting and Pre-Wiring

1. Mount the Arduino Nano onto the Screw Terminal Adapter

- Align carefully — ensure D2 matches D2, GND matches GND, etc.
- Push down gently but firmly.
- Unscrew each terminal, insert stripped wires (~0.5 cm), and tighten securely.

2. Prepare and Strip Wires

Wiring Tips:

- Strip **2 cm** of insulation from servo power wires and battery holder wires.
- Strip **3 cm** for the slide switch wires.

Use long wires for	Use short wires for
• OLED display • Push buttons • LEDs	• RTC module • Battery holder power lines

3. Connect the Power Switch

- Connect one side of the slide switch to the **positive (+)** terminal of the battery holder.
- Connect the other side to **Vin** on the Arduino Nano.
- Ensure the two sides of the switch are **not touching**.
- Use **electrical tape or hot glue** to insulate exposed metal parts and prevent shorts.

3.3 Module Wiring

(Refer to the wiring diagram for visual guidance)

Servo Motors

- **Orange** (Signal): → D5 (Servo 1), D6 (Servo 2 if using dual feeder)
- **Red**: → 5V (VCC group)
- **Brown**: → GND

Strip servo wires before connecting.

RTC Module (DS1307) *(Short wires, female jumper ends)*

- SDA → A4
- SCL → A5
- VCC → 5V
- GND → GND

Secure female connectors with hot glue.

OLED Display (SSD1306) *(Long wires, female jumper ends)*

- SDA → A4
- SCL → A5
- VCC → 5V
- GND → GND

Tip: You can twist SDA/SCL wires from the OLED and RTC together to insert into the same pins.

Push Buttons (K1 to K4) *(Long wires, female jumper ends)*

- One side of each button goes to D2, D3, D4, and D7 respectively.
- The other side connects to GND.

Ensure button pin order matches your code — incorrect wiring will switch button functions.

LEDs *(Long wires, with resistor)*

- **Long leg (anode)** → D8 (and D9 if dual), connected via 200Ω resistor.
- **Short leg (cathode)** → GND

Attach resistor to the anode, twist securely, and insulate with tape or glue.

Battery Holder (Series Wiring)

- **Red (positive)** → Slide switch
- **Black (negative)** → GND group

Ensure the holder is rewired in series (total voltage = 7.4V for 2x 18650 cells)

3.4 Power Junctions

Group and connect the following wires using screw terminals, wire nuts, or soldering.
Always insulate exposed joints with tape or glue.

5V (VCC) Group	GND Group
• Servo red wires (1 for single feeder, for 2 dual) • RTC VCC • OLED VCC • *Arduino 5V	• Servo brown wires (1 for single feeder, for 2 dual) • RTC GND • OLED GND • *Arduino GND • LED shorter leg (cathode / GND side) • Push Buttons GND • Battery pack negative (-) wire

*At the end, **connect the entire + group to Arduino's 5V**, and **- group to Arduino's GND**.

Use solder or hot glue to reinforce junctions and prevent shorts from wires touching.

3.5 Enclosure Assembly and Wire Securing

- Screw the servo arm into place (not too tightly).
- Mount the Arduino, modules, and battery holder inside the **3D-printed enclosure**.
- Use **adhesive pads, cable ties, or hot glue** to secure components.
- Leave **slack in long wires** to avoid tension on connectors.
- Keep the **OLED display and buttons accessible** at the front.
- Ensure **servo arms are not blocked** by wiring.

4. Final Inspection

Before closing the case:

Double-check:

- All wires are securely connected and not frayed or touching where they shouldn't.
- **Power switch** functions properly.
- **GND connections** are complete.
- **Servos move freely** without obstruction.
- **OLED screen** turns on.
- **Buttons** trigger input as expected.

6. Insert Assembly into CAD Enclosure

- Carefully guide all wires through the provided slots in the enclosure.
- Position the **OLED display**, **push buttons**, and **LEDs** so they align with the designated front-panel cutouts.
- Ensure the **slide switch** and **battery holder** fit snugly in their designated compartments.
- Avoid pinching or sharply bending wires. Leave some slack for future adjustments or servicing.

7. Final Setup and Uploading Code

Connect and Upload

- Connect the Arduino Nano to your computer using a USB cable.
- Open the **Arduino IDE**.

Install Required Libraries

Make sure the following libraries are installed under **Sketch > Include Library > Manage Libraries**:

- `Wire.h` — for I2C communication
- `RTClib.h` — for the RTC module
- `Adafruit_SSD1306.h` — for the OLED screen
- `Servo.h` — for servo motor control

Upload and Test

- Upload your custom sketch or one of the sample test scripts.
- Test each component **individually** before closing the case:
    - Servo movement
    - LED blinking
    - Button response
    - OLED display output
- Once all systems work as expected, **close the enclosure** securely.

8. Troubleshooting & Debugging Tips

Common Fixes

- **Loose connections:** Tug each jumper wire lightly to check for grip. Reseat if necessary.
- **Battery issues:** Ensure battery voltage is above 6.5V (for a 7.4V system). Replace cells if needed.
- **Blank screen:** Check OLED SDA/SCL wiring and I2C address.
- **Servo not moving:** Check pin assignment in code and ensure GND is shared with Arduino.
- **Buttons unresponsive:** Double-check pin numbers and ensure GND side is correctly wired.

Wiring schematic

(dashed lines are optional, only for the dual feeder)